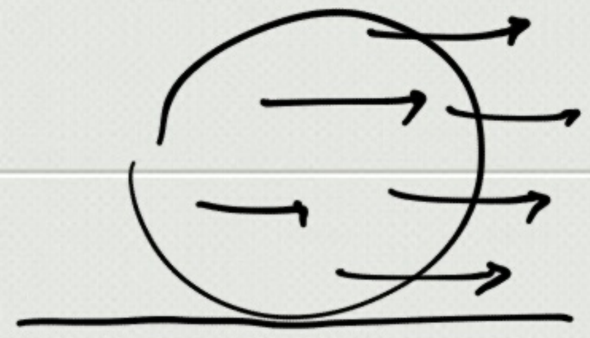
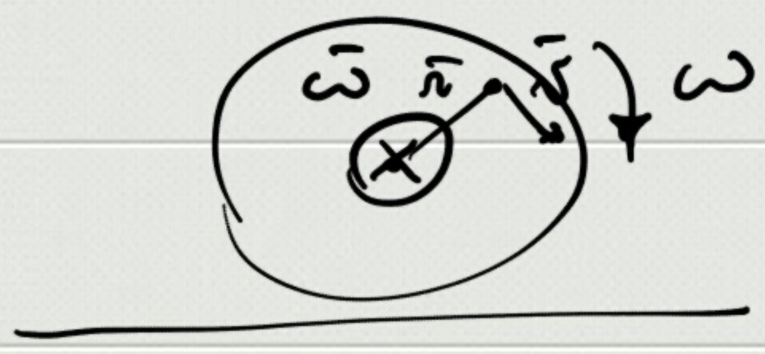
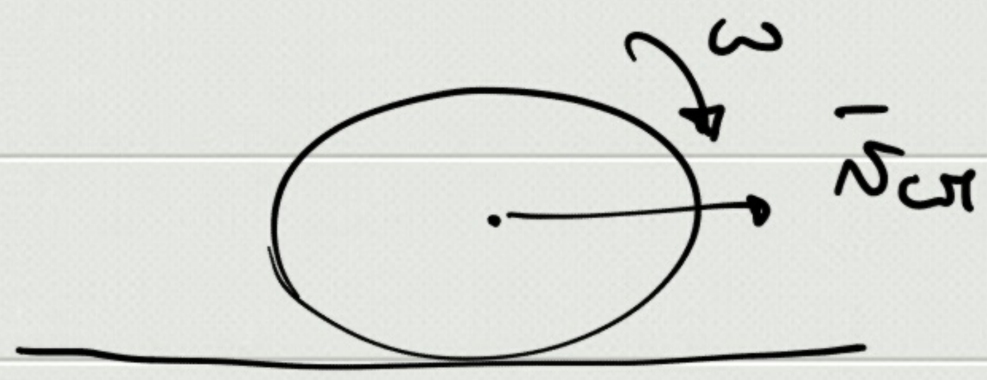


traslazione $\quad \bar{v} = \bar{v}_{CM}$

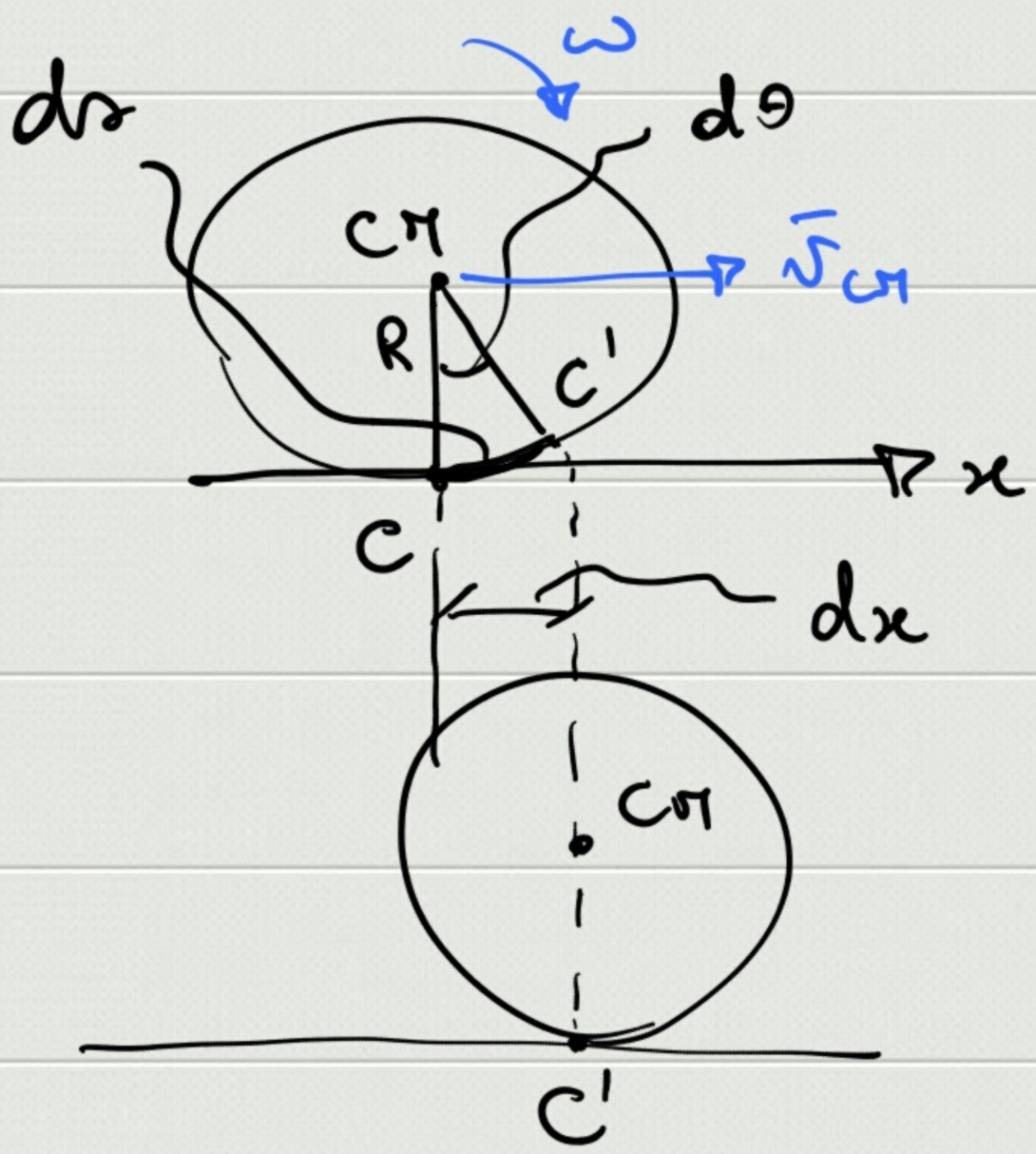


pura rotazione $\quad v_{CM} = 0$

$$\bar{v} = \bar{\omega} \times \bar{r}$$



Rototraslazione



$dt \qquad \bar{v}_{CM}, \bar{\omega}$

$$dx = v_{CM}\, dt$$

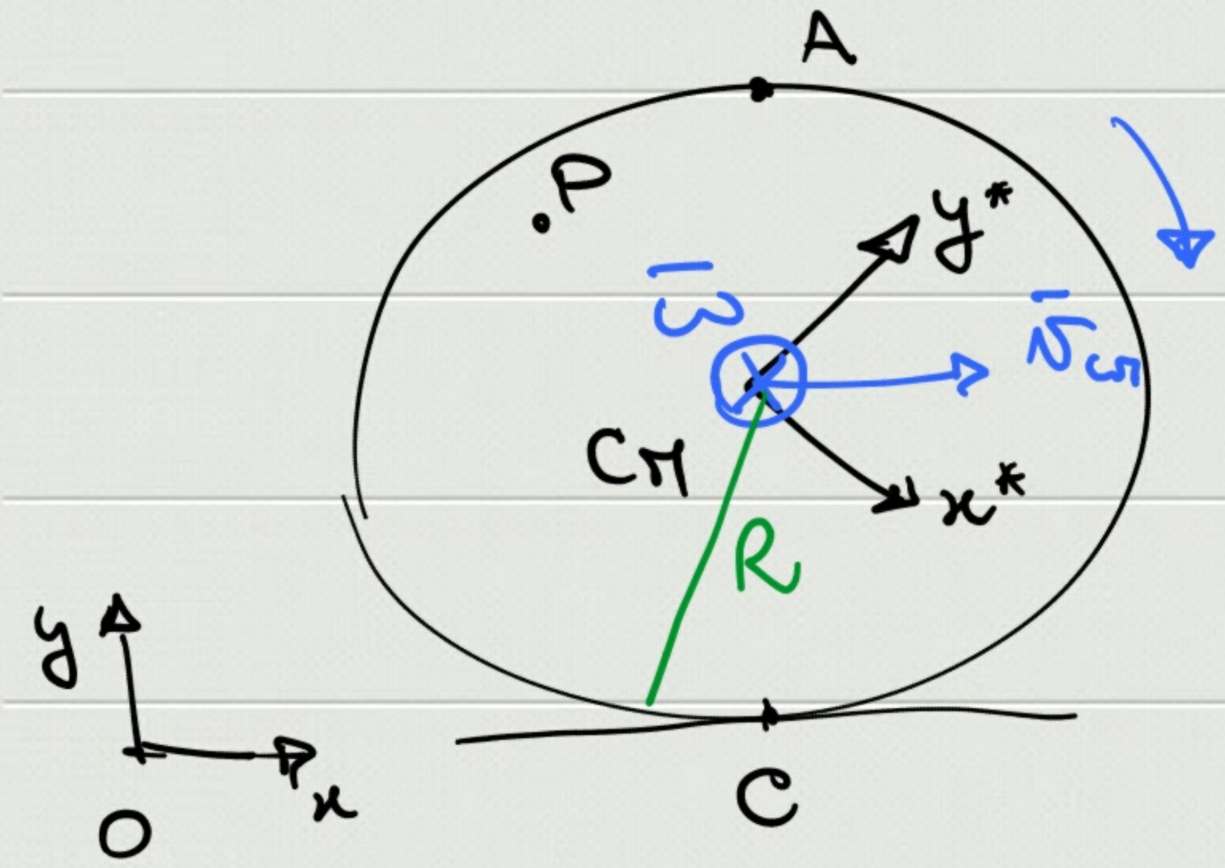
$$dx = ds = R\, d\theta$$

$$\overset{!}{=} v_{CM}\, dt$$

$$\Rightarrow v_{CM} = R \frac{d\theta}{dt} = R\omega \Rightarrow \boxed{v_{CM} = \omega R}$$

puro rotolamento

$$\boxed{v_{CM} = \omega R} \quad : \quad \frac{d}{dt} \Rightarrow \boxed{a_{CM} = \alpha R}$$



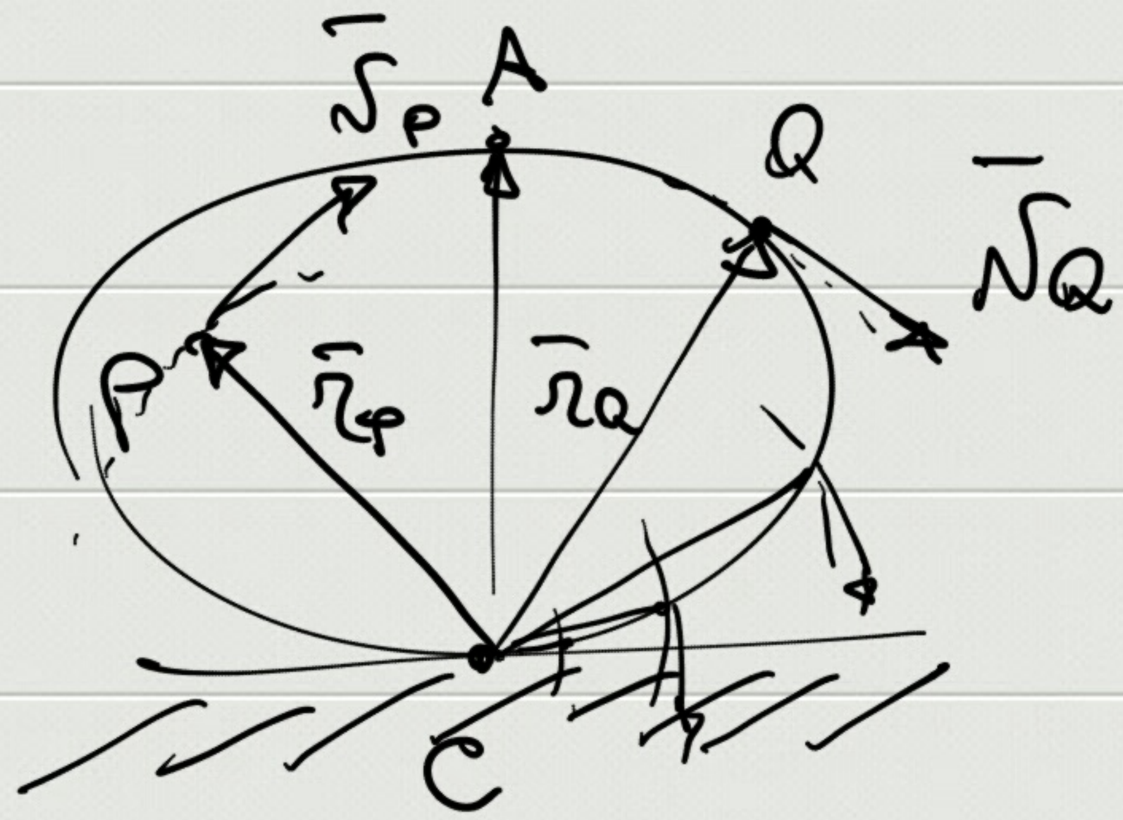
CM x^*y^* : solidale al corpo

$$\bar{v}_P = \bar{v}'_P + \bar{v}_{O'} + \bar{\omega} \times \bar{r}'_P$$

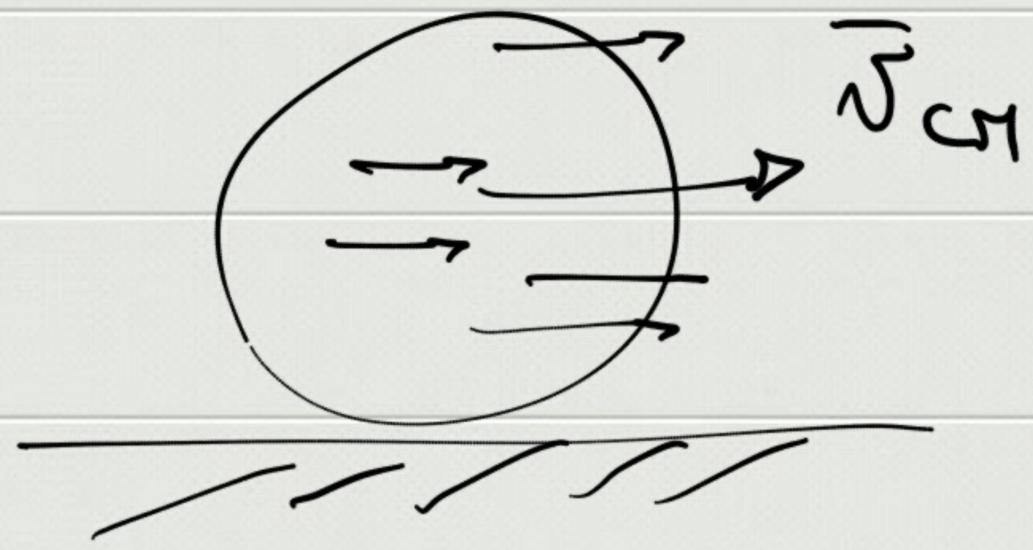
$$\Rightarrow \boxed{\bar{v}_P = \bar{v}_{CM} + \bar{\omega} \times \bar{r}'_P}$$

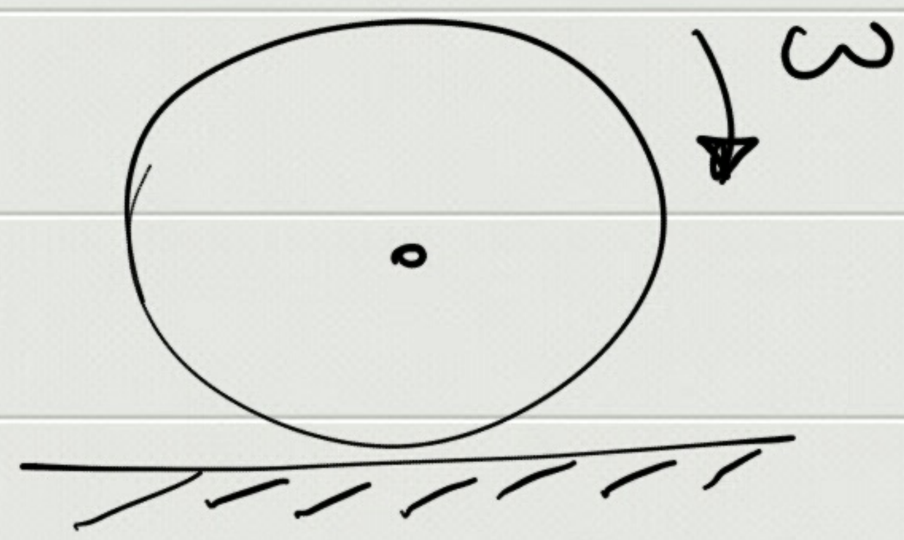
$$\bar{v}_A = \omega R \bar{u}_x + \omega R \bar{u}_x = 2\omega R \bar{u}_x = 2\bar{v}_{CM}$$

$$\bar{v}_C = \omega R \bar{u}_x + \omega R(-\bar{u}_x) = 0$$



$$\bar{v} = \bar{\omega} \times \bar{r}$$


$\vec{v}_{CM}$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